

AWS Certified AI Practitioner

AIF-C01 — Full Practice Exam

A full-length, 65-question simulation built to the official domain blueprint. Mixed tiers (Recall / Application / Analysis) and multiple-response items, mirroring the exam's scenario style. A quick-reference answer grid follows the questions, then detailed explanations for every item.

Domain	Weight	Questions
D3 — Applications of Foundation Models	28%	18
D2 — Fundamentals of Generative AI	24%	16
D1 — Fundamentals of AI & ML	20%	13
D4 — Responsible AI	14%	9
D5 — Security, Compliance & Governance	14%	9

This is original practice material — no real exam items are reproduced.

Questions

Q1 · D3 · Application

1. A retailer wants to launch a generative product-description feature. It has no ML staff, refuses to manage servers, and wants to try models from several vendors behind one API. Which service?

- A. Amazon Bedrock
- B. Amazon SageMaker Studio
- C. Amazon EC2
- D. Amazon SageMaker JumpStart

Q2 · D3 · Analysis

2. A firm needs its assistant to cite the newest internal policy, which is edited weekly, using the least development effort. Which approach?

- A. Fine-tune weekly
- B. Bedrock Knowledge Base (RAG)
- C. Raise temperature
- D. Continued pre-training

Q3 · D3 · Application

3. A chatbot must block a banned subject and redact any customer credit-card numbers it might output. Which Bedrock capability?

- A. Guardrails
- B. Clarify
- C. Agents
- D. Provisioned throughput

Q4 · D3 · Recall

4. Which signal most strongly indicates SageMaker JumpStart over Bedrock?

- A. Serverless, no ML skills
- B. Deploy an open-source FM inside our VPC
- C. Filter harmful content
- D. Find PII in S3

Q5 · D3 · Application

5. A team can supply a few example outputs and wants brand-consistent copy with minimal effort. Which technique?

- A. Few-shot prompting
- B. Zero-shot prompting
- C. Fine-tuning
- D. Continued pre-training

Q6 · D3 · Application

6. An app must pull key-value fields from thousands of scanned invoices. Which service?

- A. Amazon Textract
- B. Amazon Comprehend
- C. Amazon Rekognition
- D. Amazon Kendra

Q7 · D3 · Analysis

7. A support workflow must, in one request, call an internal order API, consult a returns document, and draft a reply. Which Bedrock capability?

- A. Bedrock Agents
- B. Guardrails
- C. Titan Embeddings
- D. On-demand pricing

Q8 · D3 · Application

8. A nightly job scores millions of records with no latency requirement. Most cost-effective inference option?

- A. Batch transform
- B. Real-time endpoint
- C. Asynchronous inference
- D. Serverless inference

Q9 · D3 · Recall

9. Amazon Kendra primarily provides what?

- A. Enterprise search that returns relevant documents
- B. A generated answer from documents
- C. Speech-to-text
- D. Image labeling

Q10 · D3 · Application

10. A media team wants natural-sounding spoken audio generated from written scripts. Which service?

- A. Amazon Polly
- B. Amazon Transcribe
- C. Amazon Lex
- D. Amazon Comprehend

Q11 · D3 · Application

11. A call center wants to convert recorded calls (with domain jargon) into accurate transcripts. Which service and feature?

- A. Amazon Transcribe with a custom language model
- B. Amazon Polly custom voice
- C. Amazon Translate custom terminology
- D. Amazon Lex intents

Q12 · D3 · Recall

12. Which pairing correctly maps Amazon Q variants?

- A. Q Developer → code/IDE; Q Business → enterprise knowledge Q&A;
- B. Q Developer → HR; Q Business → coding
- C. Both are image tools
- D. Both are databases

Q13 · D3 · Application

13. A team wants to enrich an FM with private data using the least development effort and cite sources. Which technique?

- A. RAG (Bedrock Knowledge Base)
- B. Fine-tuning

- C. In-context learning by pasting all docs
- D. Continued pre-training

Q14 · D3 · Application

14. Which Amazon OpenSearch capability makes it suitable as a vector store for RAG?

- A. Scalable index management with nearest-neighbor (k-NN) search
- B. Geospatial indexing
- C. Real-time video analysis
- D. Managed relational joins

Q15 · D3 · Application

15. A business analyst with no coding wants to build a churn predictor visually. Which SageMaker tool?

- A. SageMaker Canvas
- B. SageMaker Studio
- C. SageMaker Pipelines
- D. SageMaker Neo

Q16 · D3 · Application

16. A team wants AWS to automatically build candidate models while retaining full visibility into the steps. Which tool?

- A. SageMaker Autopilot
- B. SageMaker Canvas
- C. SageMaker Ground Truth
- D. SageMaker Feature Store

Q17 · D3 · Recall

17. Bedrock's on-demand pricing is best when the company wants what?

- A. Pay-per-token with no long-term commitment
- B. Reserved capacity for a 1-year term
- C. Free unlimited usage
- D. To train from scratch cheaply

Q18 · D3 · Application

18. A model must analyze uploaded photos to detect prohibited objects for content moderation. Which service?

- A. Amazon Rekognition
- B. Amazon Textract
- C. Amazon Comprehend
- D. Amazon Transcribe

Q19 · D2 · Recall

19. A foundation model is best described as...

- A. A large general-purpose model pre-trained on broad data, adaptable to many tasks
- B. A single-task model
- C. A deployment pipeline
- D. A labeled dataset

Q20 · D2 · Application

20. Which FM property determines how much text can be sent and returned in one request?

- A. Context window
- B. Temperature
- C. Top-p
- D. Parameter count

Q21 · D2 · Analysis

21. Which metrics best evaluate an abstractive text summarizer against references?

- A. ROUGE and BERTScore
- B. Accuracy and F1
- C. RMSE and R²
- D. AUC and log-loss

Q22 · D2 · Recall

22. In a GAN, the generator's objective is to...

- A. Produce samples realistic enough to fool the discriminator
- B. Classify real vs fake
- C. Store embeddings
- D. Reduce features

Q23 · D2 · Application

23. A model states a confident but fabricated fact. Name the phenomenon and a primary mitigation.

- A. Hallucination; ground with RAG
- B. Overfitting; regularize
- C. Drift; retrain
- D. Bias; augment data

Q24 · D2 · Recall

24. Which is an AWS first-party foundation model family?

- A. Amazon Titan
- B. Anthropic Claude
- C. Meta Llama
- D. Cohere Command

Q25 · D2 · Analysis · Select TWO

25. A company will fine-tune an FM on proprietary data. Which TWO are the main trade-offs? (Select TWO)

- A. Higher cost
- B. Higher implementation complexity
- C. Reduced accuracy
- D. Guaranteed no hallucination
- E. Lower latency

Q26 · D2 · Recall

26. Vector embeddings represent text as...

- A. Dense vectors in a space where distance encodes semantic similarity
- B. Compressed images
- C. SQL rows
- D. Plain-text keywords

Q27 · D2 · Application

27. Which parameter increases the randomness/creativity of an LLM's output?

- A. Higher temperature
- B. More epochs
- C. Larger batch size
- D. Lower learning rate

Q28 · D2 · Recall

28. Stable Diffusion is primarily used for...

- A. Generating images from text prompts
- B. Speech recognition
- C. Text classification
- D. Data compression

Q29 · D2 · Application

29. A team needs to generate SQL from natural-language questions. Which model type fits?

- A. A generative transformer (LLM)
- B. A convolutional image network
- C. A support vector machine
- D. An audio WaveNet

Q30 · D2 · Recall

30. Top-p (nucleus) sampling controls output by...

- A. Truncating to the smallest set of tokens whose cumulative probability $\geq p$
- B. Scaling the entire logit distribution
- C. Setting the token limit
- D. Selecting the model size

Q31 · D2 · Application

31. Which is a valid data format for instruction fine-tuning?

- A. Prompt–response text pairs
- B. Unlabeled images
- C. Music playlists
- D. Raw log files

Q32 · D2 · Recall

32. Transformers scale to long context primarily because of...

- A. Self-attention weighing token-to-token relationships in parallel
- B. Sequential recurrence
- C. Manual feature engineering
- D. Fixed-size lookup tables

Q33 · D2 · Application

33. A classification model is evaluated on how many predictions are correct overall. Which metric?

- A. Accuracy
- B. RMSE
- C. R^2
- D. ROUGE

Q34 · D2 · Analysis

34. Which technique ensures a generative answer reflects the most current information?

- A. RAG
- B. Fine-tuning
- C. Reinforcement learning
- D. Transfer learning

Q35 · D1 · Application

35. A model scores 97% on training data and 70% on test data. What is this?

- A. Overfitting
- B. Underfitting
- C. Class imbalance
- D. Too few features

Q36 · D1 · Recall

36. Which is a hyperparameter?

- A. Learning rate
- B. A learned weight
- C. A predicted value
- D. A learned bias

Q37 · D1 · Application

37. Which is an unsupervised learning task?

- A. Clustering customers without labels
- B. Spam classification
- C. House-price regression
- D. Sales forecasting

Q38 · D1 · Application

38. Which is a classification (not regression) problem?

- A. Labeling a transaction fraud/legit
- B. Predicting temperature
- C. Estimating revenue
- D. Predicting house price

Q39 · D1 · Application

39. To raise accuracy toward an acceptance level by giving more passes over the data, adjust...

- A. Epochs (increase)
- B. Temperature
- C. Top-p
- D. Context window

Q40 · D1 · Recall

40. Reinforcement learning's objective is to...

- A. Maximize cumulative reward over time
- B. Minimize labeled loss
- C. Cluster unlabeled data
- D. Reduce feature count

Q41 · D1 · Application · Select TWO

41. A team wants to add derived input variables AND tune training configuration to lift accuracy. Which TWO steps? (Select TWO)

- A. Feature engineering
- B. Hyperparameter tuning
- C. Model monitoring
- D. Data archival
- E. Endpoint autoscaling

Q42 · D1 · Recall

42. Data augmentation's primary purpose is to...

- A. Increase training data via transformations
- B. Reduce features
- C. Split data into folds
- D. Deploy models

Q43 · D1 · Recall

43. Which SageMaker algorithm is purpose-built for time-series forecasting?

- A. DeepAR
- B. K-Means
- C. XGBoost
- D. PCA

Q44 · D1 · Recall

44. "Label a large dataset using humans plus ML assistance" maps to...

- A. SageMaker Ground Truth
- B. Feature Store
- C. Model Monitor
- D. Neo

Q45 · D1 · Application

45. A deployed model analyzes a new image and returns a prediction. This runtime step is...

- A. Inference
- B. Training
- C. Labeling
- D. Feature engineering

Q46 · D1 · Recall

46. Dimensionality reduction (e.g., PCA) aims to...

- A. Reduce features while retaining important information
- B. Add more features
- C. Generate synthetic labels
- D. Increase epochs

Q47 · D1 · Recall

47. SageMaker Feature Store's key benefit is...

- A. Consistent, reusable features across models
- B. Faster GPUs
- C. Automatic deployment
- D. Content filtering

Q48 · D4 · Application

48. A regulated model must be transparent and its predictions explainable. Which service?

- A. SageMaker Clarify
- B. Amazon Macie
- C. Bedrock Guardrails
- D. Amazon Inspector

Q49 · D4 · Analysis

49. Which statement correctly distinguishes Guardrails from Clarify?

- A. Guardrails = runtime content filtering; Clarify = bias/explainability at training/eval
- B. Guardrails explains predictions; Clarify blocks topics
- C. Both are IAM features
- D. Both are pricing models

Q50 · D4 · Application

50. An image generator's training data over-represents one group, biasing output. Best source-level fix?

- A. Data augmentation to rebalance under-represented classes
- B. Raise temperature
- C. Batch inference
- D. More epochs

Q51 · D4 · Application

51. Only predictions below a confidence threshold should go to a human. Which service?

- A. Amazon A2I
- B. Guardrails
- C. Model Cards
- D. Comprehend

Q52 · D4 · Recall

52. SageMaker Model Cards are used to...

- A. Record intended uses, risks, training and evaluation details
- B. Serve real-time predictions
- C. Detect PII in S3
- D. Tune hyperparameters

Q53 · D4 · Application · Select TWO

53. A firm deploying an LLM responsibly should do which TWO? (Select TWO)

- A. Include fairness metrics in evaluation
- B. Mitigate bias in training data
- C. Raise temperature
- D. Avoid all logging
- E. Disable human review

Q54 · D4 · Recall

54. In AI ethics, the principle of autonomy means...

- A. Preserve human agency / keep humans in control
- B. Give AI full freedom
- C. Maximize compute

D. Minimize cost

Q55 · D4 · Application

55. A company wants to ensure its chatbot never returns inappropriate images. Best approach?

- A. Apply content-moderation (Guardrails / moderation APIs)
- B. Increase epochs
- C. Lower learning rate
- D. Add more GPUs

Q56 · D4 · Recall

56. A key acknowledged limitation of generative AI is that it...

- A. May produce biased or inappropriate content requiring human review
- B. Cannot handle multiple languages
- C. Cannot process large data
- D. Cannot generate code

Q57 · D5 · Application

57. A security team needs a log of every Bedrock API call — who called what, when — to probe unauthorized access. Which service?

- A. AWS CloudTrail
- B. Amazon CloudWatch
- C. Amazon Inspector
- D. AWS Trusted Advisor

Q58 · D5 · Application

58. A company must discover and classify PII sitting in S3 buckets. Which service?

- A. Amazon Macie
- B. Amazon Inspector
- C. AWS Config
- D. AWS Artifact

Q59 · D5 · Application

59. A team must scan EC2 and ECR for known vulnerabilities. Which service?

- A. Amazon Inspector
- B. Amazon Macie
- C. AWS CloudTrail
- D. AWS Artifact

Q60 · D5 · Analysis

60. Using the generative-AI security scoping model, which gives the customer the MOST security responsibility?

- A. Training a model from scratch on owned data
- B. Using embedded-GenAI SaaS
- C. Building on a third-party FM API
- D. Fine-tuning a third-party FM

Q61 · D5 · Application

61. To control which principals may invoke Bedrock, following least privilege, use...

- A. AWS IAM

- B. Amazon Macie
- C. Amazon CloudWatch
- D. AWS X-Ray

Q62 · D5 · Recall

62. AWS Artifact provides...

- A. On-demand compliance reports and agreements (SOC, ISO, PCI)
- B. Vulnerability scans
- C. API call logs
- D. Model bias reports

Q63 · D5 · Application · Select TWO

63. A chatbot must demonstrate regulatory compliance. Which TWO capabilities are most relevant? (Select TWO)

- A. Threat detection
- B. Data protection
- C. Cost optimization
- D. Autoscaling
- E. Loose coupling

Q64 · D5 · Application

64. Which action best hardens an LLM against prompt-injection attacks?

- A. A system prompt/template that detects and refuses manipulation, plus Guardrails prompt-attack filtering
- B. Raise temperature
- C. Reduce input tokens
- D. Use only SageMaker-listed models

Q65 · D5 · Recall

65. Which service provides end-to-end distributed request tracing across services?

- A. AWS X-Ray
- B. AWS CloudTrail
- C. Amazon CloudWatch
- D. AWS Config

Answer grid

#	Ans	#	Ans	#	Ans	#	Ans	#	Ans
1	A	14	A	27	A	40	A	53	A, B
2	B	15	A	28	A	41	A, B	54	A
3	A	16	A	29	A	42	A	55	A
4	B	17	A	30	A	43	A	56	A
5	A	18	A	31	A	44	A	57	A
6	A	19	A	32	A	45	A	58	A
7	A	20	A	33	A	46	A	59	A
8	A	21	A	34	A	47	A	60	A
9	A	22	A	35	A	48	A	61	A
10	A	23	A	36	A	49	A	62	A
11	A	24	A	37	A	50	A	63	A, B
12	A	25	A, B	38	A	51	A	64	A
13	A	26	A	39	A	52	A	65	A

Score yourself: 700/1000 is roughly 70%+. Below that, revisit the domains where you missed most — weight review toward D3 and D2 (over half the exam).

Explanations

Q1 — Correct: A [D3 · Application]

Why. No ML staff + no server management + one multi-vendor API = Amazon Bedrock (serverless FM access).

Why the others are wrong. **B** — Studio is full custom build — maximum ops. **C** — Self-hosting is the most overhead. **D** — JumpStart still hands you an endpoint to run.

Q2 — Correct: B [D3 · Analysis]

Why. Fresh, frequently changing facts with least effort = RAG via a managed Knowledge Base (retrieval at inference, no retraining).

Why the others are wrong. **A** — Weekly fine-tuning is heavy and lags data. **C** — Temperature is randomness, not knowledge. **D** — Pre-training is the heaviest option.

Q3 — Correct: A [D3 · Application]

Why. Denied topics + PII redaction on inputs/outputs at runtime = Bedrock Guardrails.

Why the others are wrong. **B** — Clarify is training-time bias/explainability. **C** — Agents orchestrate tasks. **D** — A pricing option.

Q4 — Correct: B [D3 · Recall]

Why. Open-source model owned and deployed in your VPC = JumpStart (you own the endpoint).

Why the others are wrong. **A** — That's Bedrock. **C** — That's Guardrails. **D** — That's Macie.

Q5 — Correct: A [D3 · Application]

Why. A few in-prompt examples + least effort + style = few-shot.

Why the others are wrong. **B** — Zero-shot gives no examples. **C** — Fine-tuning needs many labeled pairs. **D** — Pre-training is heaviest.

Q6 — Correct: A [D3 · Application]

Why. OCR + form/key-value extraction from documents = Textract.

Why the others are wrong. **B** — Comprehend analyzes meaning, not OCR. **C** — Rekognition is images/video. **D** — Kendra is enterprise search.

Q7 — Correct: A [D3 · Analysis]

Why. Multi-step orchestration across APIs/knowledge = Bedrock Agents.

Why the others are wrong. **B** — Guardrails filters content. **C** — Embeddings support retrieval. **D** — A pricing model.

Q8 — Correct: A [D3 · Application]

Why. Offline bulk, no urgency = Batch Transform.

Why the others are wrong. **B** — Real-time is for live low-latency. **C** — Async is for large payloads within ~an hour. **D** — Serverless is intermittent real-time.

Q9 — Correct: A [D3 · Recall]

Why. Kendra is managed enterprise search returning documents/passages.

Why the others are wrong. **B** — That's a Bedrock Knowledge Base (RAG). **C** — That's Transcribe. **D** — That's Rekognition.

Q10 — Correct: A [D3 · Application]

Why. Text-to-speech with natural voices = Polly.

Why the others are wrong. **B** — Transcribe is speech-to-text. **C** — Lex builds chatbots. **D** — Comprehend is NLP analysis.

Q11 — Correct: A [D3 · Application]

Why. Speech-to-text with domain vocabulary = Transcribe custom language model.

Why the others are wrong. **B** — Polly is TTS. **C** — Translate is language translation. **D** — Lex is conversational bots.

Q12 — Correct: A [D3 · Recall]

Why. Q Developer works in code/IDE; Q Business answers over enterprise knowledge.

Why the others are wrong. **B** — Reversed/incorrect. **C** — Neither is an image tool. **D** — Neither is a database.

Q13 — Correct: A [D3 · Application]

Why. External knowledge + least effort + citations = RAG.

Why the others are wrong. **B** — Fine-tuning bakes knowledge, higher effort, no live citations. **C** — Context window limits pasting everything. **D** — Heaviest option.

Q14 — Correct: A [D3 · Application]

Why. Vector similarity retrieval relies on k-NN nearest-neighbor search over an index.

Why the others are wrong. **B** — Geospatial is a different feature. **C** — Not an OpenSearch function. **D** — Not the vector-search capability.

Q15 — Correct: A [D3 · Application]

Why. No-code, visual model building = Canvas.

Why the others are wrong. **B** — Studio is the code IDE. **C** — Pipelines orchestrate workflows. **D** — Neo compiles for edge devices.

Q16 — Correct: A [D3 · Application]

Why. “Automatically” + “full visibility” = Autopilot (AutoML that exposes its pipeline).

Why the others are wrong. **B** — Canvas hides complexity (no-code). **C** — Ground Truth labels data. **D** — Feature Store reuses features.

Q17 — Correct: A [D3 · Recall]

Why. On-demand = per-token, no commitment, ideal for flexible/limited budgets.

Why the others are wrong. **B** — That's provisioned throughput. **C** — No such offering. **D** — Unrelated to on-demand inference pricing.

Q18 — Correct: A [D3 · Application]

Why. Object detection and moderation on images = Rekognition.

Why the others are wrong. **B** — Textract reads text in documents. **C** — Comprehend is text NLP. **D** — Transcribe is speech-to-text.

Q19 — Correct: A [D2 · Recall]

Why. Broad pre-training + general adaptability defines an FM.

Why the others are wrong. **B** — That's a narrow model. **C** — Not a model. **D** — Data, not a model.

Q20 — Correct: A [D2 · Application]

Why. Context window = max tokens (input+output) per request.

Why the others are wrong. **B** — Randomness control. **C** — Sampling diversity control. **D** — Capability, not per-request capacity.

Q21 — Correct: A [D2 · Analysis]

Why. Text generation → overlap (ROUGE) + semantic similarity (BERTScore).

Why the others are wrong. **B** — Classification metrics. **C** — Regression metrics. **D** — Classification-probability metrics.

Q22 — Correct: A [D2 · Recall]

Why. The generator creates fakes to fool the discriminator; the competition improves both.

Why the others are wrong. **B** — That's the discriminator. **C** — Not its role. **D** — Unrelated.

Q23 — Correct: A [D2 · Application]

Why. Confident fabrication = hallucination; grounding via RAG is a primary mitigation.

Why the others are wrong. **B** — Training-generalization issue. **C** — Distribution shift over time. **D** — Fairness issue, different problem.

Q24 — Correct: A [D2 · Recall]

Why. Titan is AWS's own FM family on Bedrock.

Why the others are wrong. **B** — Anthropic. **C** — Meta. **D** — Cohere.

Q25 — Correct: A, B [D2 · Analysis]

Why. Customization buys fit at the cost of training expense and engineering complexity.

Why the others are wrong. **C** — Accuracy should improve. **D** — Not guaranteed. **E** — Not the trade-off.

Q26 — Correct: A [D2 · Recall]

Why. Embeddings map meaning to geometry, enabling similarity search/RAG.

Why the others are wrong. **B** — Not images. **C** — Not rows. **D** — Not keywords.

Q27 — Correct: A [D2 · Application]

Why. Temperature scales the output distribution — higher = flatter = more random.

Why the others are wrong. **B** — Training setting. **C** — Training setting. **D** — Training setting.

Q28 — Correct: A [D2 · Recall]

Why. Text-to-image diffusion generation.

Why the others are wrong. **B** — That's Transcribe/ASR. **C** — NLP task. **D** — Unrelated.

Q29 — Correct: A [D2 · Application]

Why. Text-to-text generation is an LLM/transformer task.

Why the others are wrong. **B** — For images. **C** — Traditional classifier. **D** — Audio generation.

Q30 — Correct: A [D2 · Recall]

Why. Top-p keeps the nucleus of most-probable tokens up to cumulative probability p.

Why the others are wrong. **B** — That's temperature. **C** — That's max tokens. **D** — Unrelated.

Q31 — Correct: A [D2 · Application]

Why. Instruction fine-tuning uses prompt→response (instruction) pairs.

Why the others are wrong. **B** — Not instruction pairs. **C** — Irrelevant. **D** — Irrelevant.

Q32 — Correct: A [D2 · Recall]

Why. Self-attention captures long-range dependencies in parallel, unlike RNNs.

Why the others are wrong. **B** — That's older RNN behavior. **C** — Not the mechanism. **D** — Not the mechanism.

Q33 — Correct: A [D2 · Application]

Why. Correct/total for classification = accuracy.

Why the others are wrong. **B** — Regression. **C** — Regression. **D** — Text generation.

Q34 — Correct: A [D2 · Analysis]

Why. RAG retrieves current data at inference; fine-tuning freezes knowledge at training time.

Why the others are wrong. **B** — Static once trained. **C** — Not about freshness. **D** — Not about freshness.

Q35 — Correct: A [D1 · Application]

Why. High train / low test = overfitting.

Why the others are wrong. **B** — Underfitting is low on both. **C** — Different problem. **D** — Would tend to underfit.

Q36 — Correct: A [D1 · Recall]

Why. Hyperparameters (e.g., learning rate) are set before training.

Why the others are wrong. **B** — Learned parameter. **C** — Output. **D** — Learned parameter.

Q37 — Correct: A [D1 · Application]

Why. Clustering unlabeled data is unsupervised.

Why the others are wrong. **B** — Supervised classification. **C** — Supervised regression. **D** — Supervised.

Q38 — Correct: A [D1 · Application]

Why. Category output = classification.

Why the others are wrong. **B** — Numeric = regression. **C** — Numeric = regression. **D** — Numeric = regression.

Q39 — Correct: A [D1 · Application]

Why. An epoch is one full training pass; more epochs = more learning (up to overfitting).

Why the others are wrong. **B** — Inference parameter. **C** — Inference parameter. **D** — Inference capacity.

Q40 — Correct: A [D1 · Recall]

Why. RL maximizes cumulative reward through interaction with an environment.

Why the others are wrong. **B** — Supervised optimization. **C** — Unsupervised. **D** — Dimensionality reduction.

Q41 — Correct: A, B [D1 · Application]

Why. New variables = feature engineering; training-config search = hyperparameter tuning.

Why the others are wrong. **C** — Post-deployment watch. **D** — Storage, not accuracy. **E** — Serving, not accuracy.

Q42 — Correct: A [D1 · Recall]

Why. Augmentation synthetically expands data (and rebalances classes, curbs overfitting).

Why the others are wrong. **B** — Dimensionality reduction. **C** — Cross-validation. **D** — Deployment.

Q43 — Correct: A [D1 · Recall]

Why. DeepAR is the forecasting algorithm.

Why the others are wrong. **B** — Clustering. **C** — Boosted trees. **D** — Dimensionality reduction.

Q44 — Correct: A [D1 · Recall]

Why. Ground Truth combines human + ML-assisted labeling.

Why the others are wrong. **B** — Feature reuse. **C** — Drift monitoring. **D** — Edge compilation.

Q45 — Correct: A [D1 · Application]

Why. Using a trained model on new data = inference.

Why the others are wrong. **B** — Learning phase. **C** — Data prep. **D** — Data prep.

Q46 — Correct: A [D1 · Recall]

Why. Cut features while preserving structure — fights the curse of dimensionality.

Why the others are wrong. **B** — Opposite. **C** — Not its purpose. **D** — Training setting.

Q47 — Correct: A [D1 · Recall]

Why. A shared, consistent feature repository reused across models.

Why the others are wrong. **B** — Not its role. **C** — Not its role. **D** — That's Guardrails.

Q48 — Correct: A [D4 · Application]

Why. Clarify provides bias detection and explainability (e.g., SHAP).

Why the others are wrong. **B** — Data discovery. **C** — Content filtering. **D** — Vulnerability scanning.

Q49 — Correct: A [D4 · Analysis]

Why. Different layers: content safety vs statistical fairness/transparency.

Why the others are wrong. **B** — Roles reversed. **C** — Neither is IAM. **D** — Neither is pricing.

Q50 — Correct: A [D4 · Application]

Why. Bias enters via imbalanced data; rebalance/augment under-represented classes.

Why the others are wrong. **B** — Randomness, not fairness. **C** — Irrelevant. **D** — Can entrench bias.

Q51 — Correct: A [D4 · Application]

Why. A2I routes low-confidence predictions to human review.

Why the others are wrong. **B** — Auto-blocks content. **C** — Documentation. **D** — NLP service.

Q52 — Correct: A [D4 · Recall]

Why. Immutable governance documentation of a model.

Why the others are wrong. **B** — That's an endpoint. **C** — That's Macie. **D** — That's HPO/Autopilot.

Q53 — Correct: A, B [D4 · Application]

Why. Responsible deployment = fairness metrics + bias mitigation in data.

Why the others are wrong. **C** — Randomness, unrelated. **D** — Logging aids governance. **E** — Human oversight is desirable.

Q54 — Correct: A [D4 · Recall]

Why. Autonomy here is human autonomy — humans stay in control.

Why the others are wrong. **B** — Opposite meaning. **C** — Unrelated. **D** — Unrelated.

Q55 — Correct: A [D4 · Application]

Why. A content-moderation/safety layer screens outputs.

Why the others are wrong. **B** — Training setting. **C** — Training setting. **D** — Infrastructure.

Q56 — Correct: A [D4 · Recall]

Why. Probabilistic, ungrounded output can be biased/inappropriate — needs oversight.

Why the others are wrong. **B** — Modern LLMs handle many languages. **C** — They handle large data. **D** — They generate code.

Q57 — Correct: A [D5 · Application]

Why. API audit logging (who/what/when) = CloudTrail.

Why the others are wrong. **B** — Metrics/alarms, not who-did-what. **C** — Vulnerability scanning. **D** — Best-practice checks.

Q58 — Correct: A [D5 · Application]

Why. Sensitive-data discovery/classification in S3 = Macie.

Why the others are wrong. **B** — Infra/code vulnerabilities. **C** — Config change tracking. **D** — Compliance reports.

Q59 — Correct: A [D5 · Application]

Why. Vulnerability scanning of infrastructure/containers = Inspector.

Why the others are wrong. **B** — Data discovery. **C** — API logging. **D** — Compliance reports.

Q60 — Correct: A [D5 · Analysis]

Why. Responsibility scales with ownership; training from scratch is maximum.

Why the others are wrong. **B** — Least responsibility. **C** — Less than fine-tuning. **D** — Middle of the range.

Q61 — Correct: A [D5 · Application]

Why. Authorization with least privilege = IAM.

Why the others are wrong. **B** — Data discovery. **C** — Monitoring. **D** — Tracing.

Q62 — Correct: A [D5 · Recall]

Why. Artifact is the compliance-report/attestation portal.

Why the others are wrong. **B** — That's Inspector. **C** — That's CloudTrail. **D** — That's Clarify.

Q63 — Correct: A, B [D5 · Application]

Why. Regulatory frameworks emphasize security (threat detection) and privacy (data protection).

Why the others are wrong. **C** — Operational, not compliance. **D** — Operational. **E** — Architectural.

Q64 — Correct: A [D5 · Application]

Why. Instruction hardening + Guardrails prompt-attack detection defends against injection.

Why the others are wrong. **B** — Increases susceptibility. **C** — Doesn't address injection. **D** — Irrelevant.

Q65 — Correct: A [D5 · Recall]

Why. X-Ray traces a request across microservices.

Why the others are wrong. **B** — API logging. **C** — Metrics/alarms. **D** — Config history.